

Modeling Neutralization in Exemplar Theory

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What is Neutralization?

- ▶ When a phonological contrast in a language is lost in certain contexts.
- ▶ Example cases of neutralization:
 - Obstruent Voicing in Russian (Matsui, 2015)
 - Vowel length in Japanese (Braver, 2019)
 - Vowel nasality in Mankiyali (Paramore, 2025)
 - Manner of articulation in Korean (Kim & Jongman, 1996).
- ▶ The phonetic realization of phonological neutralization can be **complete** or **incomplete**.



Phonetically Incomplete Neutralization

- ▶ Residual phonetic traces of the underlying phonological contrast persist in the acoustic record.

i.	/ki-mo/	→	[kimo]	tree-PART	μ duration of [i]: 50 ms
ii.	/ki:/	→	[ki:]	key	μ duration of [i:]: 157 ms
iii.	/ki/	→	[ki:]	tree	μ duration of [i:]: 125 ms

Table 1: *Vowel Lengthening in Japanese.* (Braver, 2019)

Phonetically Complete Neutralization



- ▶ No measurable surface differences in the neutralized contrast.

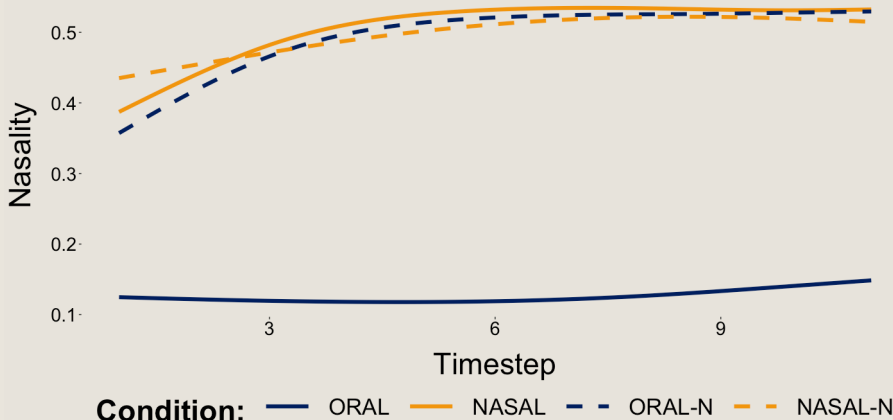


Figure 1: *Vowel Nasality Neutralization in Mankiyali.* (Paramore, 2025)

This Paper



► Research Question:

- Within an eliminativist exemplar-theoretic framework, is it predicted that we should observe both types of neutralization?

► Computational model of exemplar accumulations:

- Naturally predicts either incomplete or complete neutralization.
- Barring unnatural token frequency effects, both types of neutralization are difficult to generate.



What is Exemplar Theory?

- ▶ A cluster of theories with the common view that language use is the main driver of linguistic cognition (Bybee, 2001; Goldinger, 1998; Goldrick & Cole, 2023; Johnson, 1996, 2006; Kaplan, 2015; Nosofsky, 1986; Pierrehumbert, 2001, 2016).
- ▶ The assumptions followed here are based closely on Bybee (2001), Pierrehumbert (2001), and Kaplan (2015).

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4. Token *frequency* impacts representation. More frequent exemplars are strengthened in memory (Bybee, 1985, p.119-123).



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5. *Production biases* can shift exemplar categories over time (Pierrehumbert, 2001, p.146-148).

Modeling Neutralization



- ▶ *Model Overview*: Simulation of vowel nasality neutralization over time for an individual speaker.
- ▶ *Model components*:
 - **input**: four categories consisting of lexical items with pre-specified frequencies.
 - **exemplar accumulation**: iteratively creates new optimized exemplars based on eliminativist exemplar principles.
 - **output**: plot showing evolution of the four categories's nasality values over 1,000 iterations.

Model Input



- ▶ Four categories with two lexical items each.
 1. **ORAL**: [taa], [kii]
 2. **NASAL**: [tãã], [kĩĩ]
 3. **ORAL-N**: [taa-n], [kii-n]
 4. **NASAL-N**: [tãã-n], [kĩĩ-n]
- ▶ Each lexical item seeded with a single exemplar.
 - Content corresponds to vowel nasality
 - Nasality of **ORAL** and **ORAL-N** vowels ~ 0.2 (sd = .015)
 - Nasality of **NASAL** and **NASAL-N** vowels ~ 0.6 (sd = .015)
- ▶ Frequencies of each category in relation to other categories.

Exemplar Accumulation



Step 1: selection

Randomly select 1 of the
8 lexical items

taa
taa-n

kii
kii-n

tǎǎ
tǎǎ-n

kīī
kīī-n

Exemplar Accumulation



Step 1: selection

Randomly select 1 of the
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Step 2: entrenchment

Set nasality to mean of
item's cloud



Exemplar Accumulation



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Step 2: entrenchment

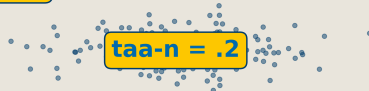
Set nasality to mean of
item's cloud



Step 3: optimization

Optimize nasality based
on exemplar biases

taa kii tãã kīī
taa-n kii-n tãã-n kīī-n



1. pre-N production bias: $5 \times (\text{NASAL} - \text{taa-n})^2$
2. morphological bias: $\text{FREQ} \times (\text{ORAL} - \text{taa-n})^2$

taa-n = .275



Exemplar Accumulation

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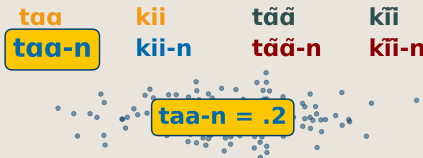
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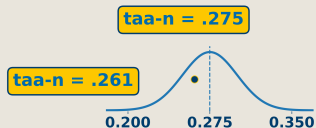
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Step 4: noise

Add random noise to the
nasality value



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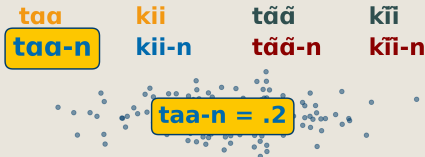
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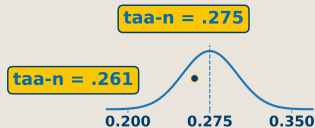


Step 5: storage

Store the new exemplar



1. pre-N production bias: $5 \times (\text{NASAL} - \text{taa-n})^2$
2. morphological bias: $\text{FREQ} \times (\text{ORAL} - \text{taa-n})^2$

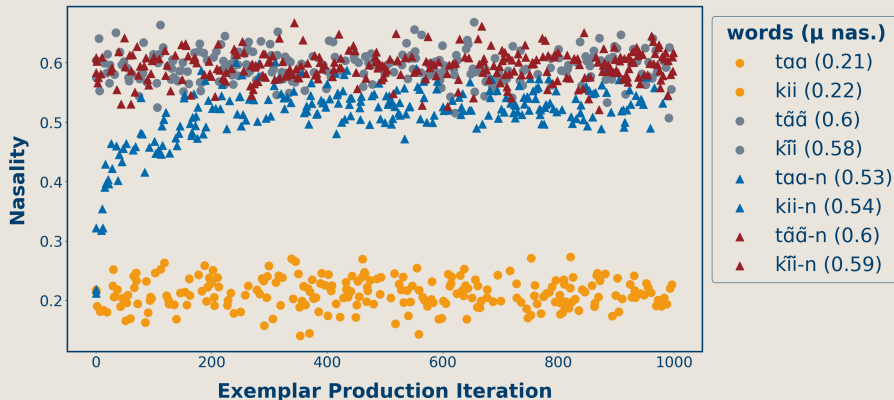


taa-n : [.18, .22, **.261**]



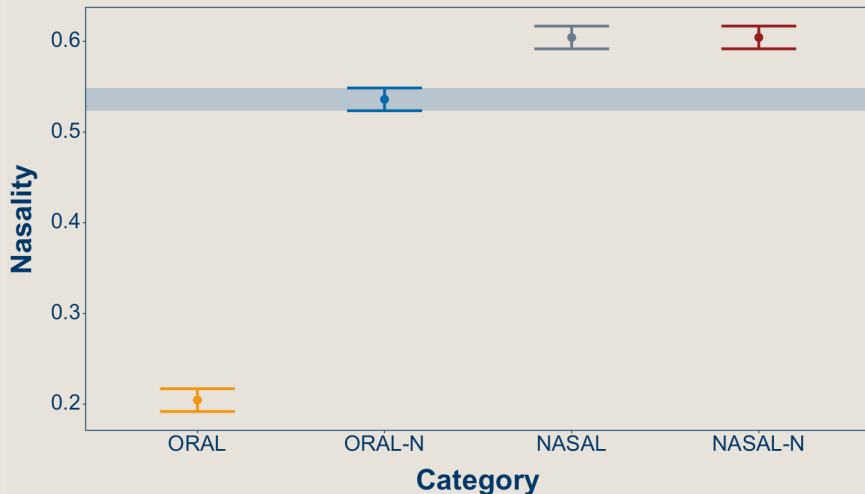
Modeling Incomplete Neutralization

Exemplar Changes in Nasality Over Time, Simulation 1





Estimates over 100 Simulations





Frequency-Driven Complete Neutralization

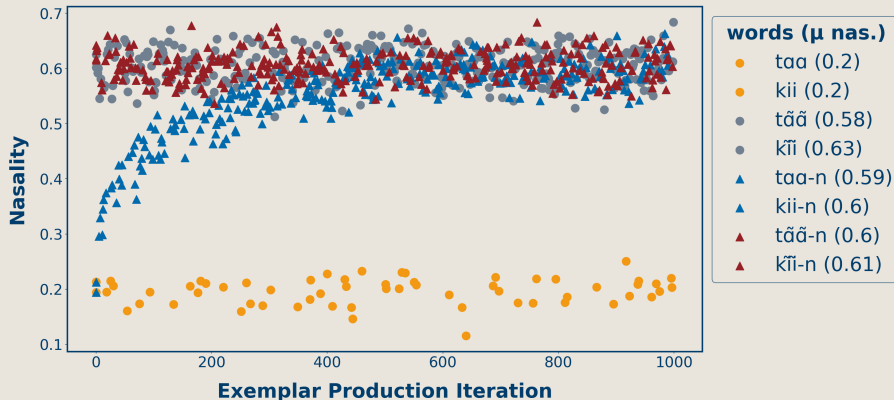
- ▶ Lowering the token frequency of base ORAL forms.
 - Morphological similarity becomes increasingly less important.
 - Only requires changes to the exemplar space.

$$FREQ \times (BASE - exemplar)^2$$



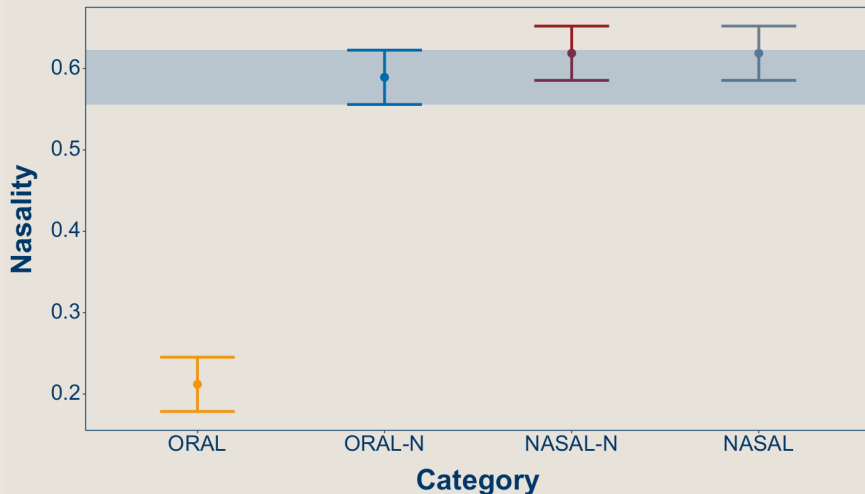
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Exemplar Changes in Nasality Over Time, Simulation 1





Frequency-Driven Complete Neutralization





Frequency-Driven Complete Neutralization

- ▶ Manipulating base token frequency may not be empirically justified.
 - Not likely that BASE forms are systematically less frequent than paradigmatically related forms in a language.
- ▶ One (potentially) questionable prediction of this approach:
 - All languages that exhibit complete neutralization have relatively low-frequency BASE forms.
 - All languages that exhibit incomplete neutralization have relatively high-frequency BASE forms.



Modeling Complete Neutralization

► My Proposal:

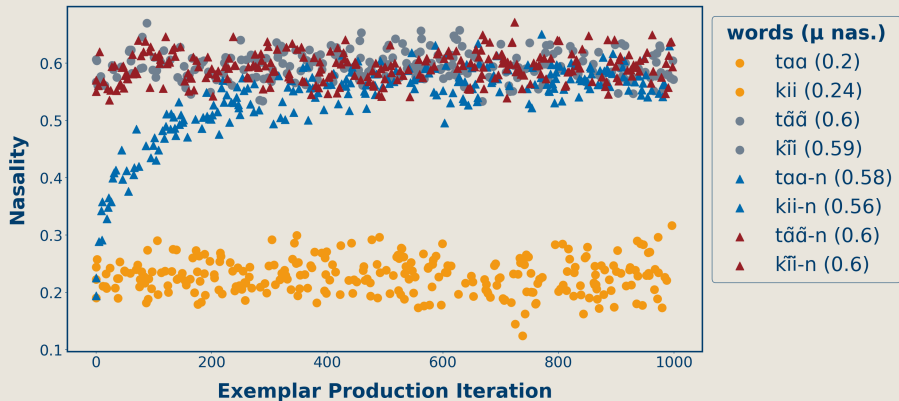
- Variation in planned output targets leads to distinct neutralization patterns.
 - Coarticulatory reanalysis leads a speaker to change the *planned* degree of nasality on ORAL-N forms (Beddor, 2009; Ohala, 1993).
- Can be implemented by changing the scalar of the pre-N nasality production bias.

$$5 \times (NASAL - exemplar)^2$$

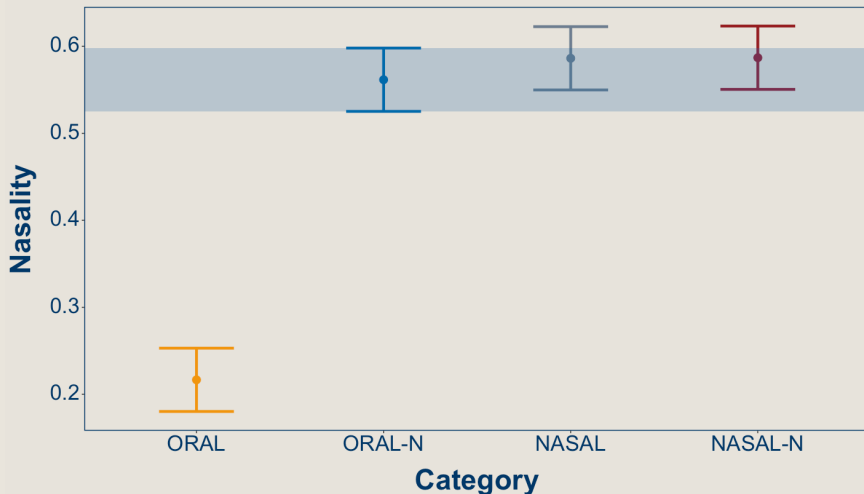


Nasality Bias: $15 \times (NASAL - exemplar)^2$

Exemplar Changes in Nasality Over Time, Simulation 1



Nasality Bias: $15 \times (NASAL - exemplar)^2$



Discussion



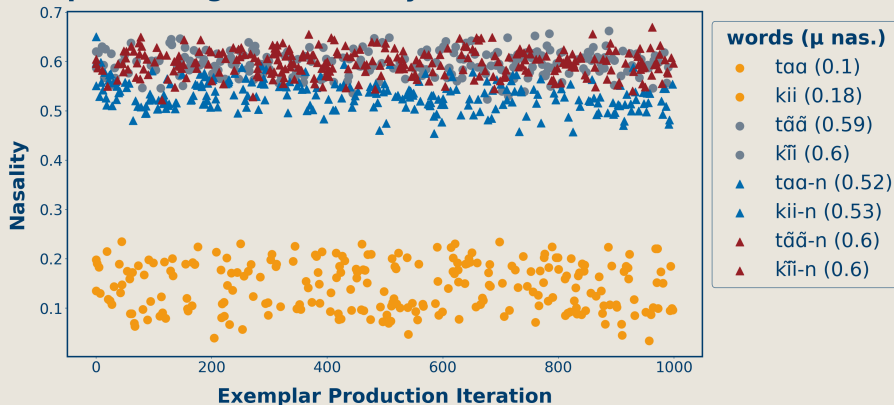
- ▶ Barring unnatural frequency effects, changes in exemplar patterns struggle to capture both types of neutralization.
- ▶ Analyzing variation as differences in planned target outputs provides a possible solution.
 - This requires planned linguistic knowledge.
 - Changes in exemplar values alone do not change the endpoint of optimization.



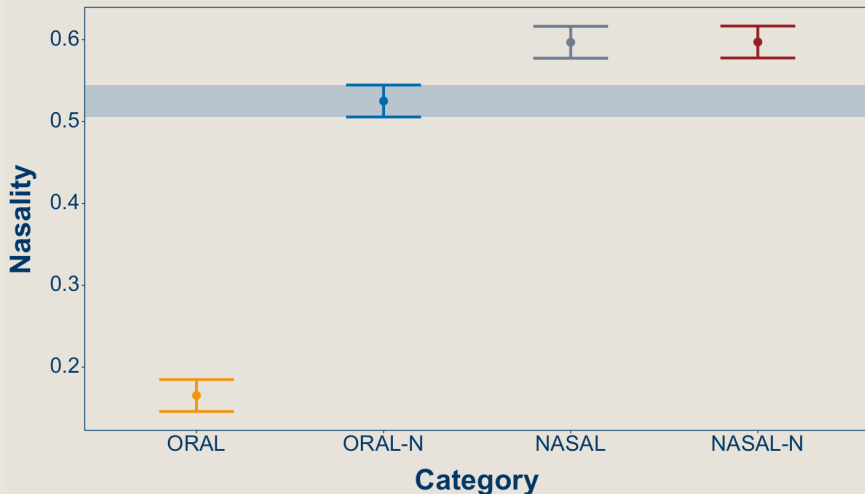
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- **taa-n** and **kii-n** initialized at 0.6

Exemplar Changes in Nasality Over Time, Simulation 1



Nasality Bias: $5 \times (NASAL - exemplar)^2$



Future Directions



- ▶ The model presented here only considers production exemplars.
 - It's well-known that production and perception are linked and both play roles in shaping the lexicon.
 - Future updates to the model should incorporate perception exemplars (Beddor et al., 2018).
- ▶ The model weights all exemplars equally.
 - Social status, recency, etc. may result in certain exemplars having a stronger influence than others on future productions.

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